

PAPER_957: Cooper Pair Lagrangian Variational Principle

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Abstract

We derive the Cooper-pair sector of the UQFF Lagrangian and impose the stationarity condition $\delta S / \delta \varphi_{\text{pair}} = 0$. The gap Lagrangian density $\mathcal{L}_{\text{gap}} = -N(0)|\Delta|^2/V_{\text{SCm}} + N(0)\hbar\omega_{\text{SCm}} \ln(2 \cosh(\Delta/2k_B T))$ yields the self-consistent BCS gap equation when varied. Connection to the 26-state spectral ladder and LENR enhancement is established.

1. Gap Lagrangian

$$\mathcal{L}_{\text{gap}} = -\frac{N(0)|\Delta|^2}{V_{\text{SCm}}} + N(0)\hbar\omega_{\text{SCm}} \ln\left(2 \cosh \frac{\Delta}{2k_B T}\right)$$

2. Stationarity Condition

$$\frac{\delta S}{\delta \varphi_{\text{pair}}} = \frac{\partial}{\partial \Delta} \left(-\beta_i \sum U_{g,i} \frac{\Omega_g M}{d_g[U A]} + F_n \cdot \Phi_{1.25\text{THz}} \right) = 0$$

This yields the self-consistent gap equation:

$$1 = \frac{V_{\text{SCm}}}{2} \cdot \frac{\tanh(\Delta/2k_B T)}{\Delta} \cdot S_{26}$$

3. SCm Gap Equation

$$\Delta = \frac{\hbar\omega_{\text{SCm}}}{2} \cdot \tanh\left(\frac{\Delta}{2k_B T}\right) \cdot S_{26} \cdot \frac{F_{UBi}}{F_U}$$

Critical temperature:

$$T_c = \frac{1.13 \hbar\omega_{\text{SCm}}}{k_B} \cdot \exp\left(-\frac{1}{N(0)V_{\text{SCm}}}\right)$$

4. Spectral Ladder Link

$$E_n = E_0 \cdot (2\pi)^{n/3} \cdot S_{26}, \quad n = 1, \dots, 26$$

The gap Δ couples to each spectral ladder level, with the phonon frequency ω_{SCm} setting the base energy $E_0 = \hbar\omega_{\text{SCm}}$.

5. LENR Connection

$$\Gamma_{\text{LENR}} \propto \Delta^2 \cdot \exp\left(-\frac{E_{\text{Coulomb}}}{k_B T_c}\right) \cdot \Phi_{1.25\text{THz}}$$

6. Source Data

- **File:** uqff_lagrangian_derivation.py §10
 - **Session:** 214
 - **CP4 Class:** CooperPairLagrangianCalc (#541)
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References

1. Bardeen, Cooper, Schrieffer – Theory of Superconductivity (1957)
2. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)